

Feature	Ethernet Cable	Fiber Optic Cable
Transmission medium	Uses electrical signals sent through copper wires.	Uses pulses of light sent through thin glass or plastic fibers.
Speed & distance	Good for short distances (typically up to 100 m) with high speeds suitable for homes and offices.	Supports much higher speeds over much longer distances (from hundreds of meters to many kilometers) with minimal signal loss.
Cost & installation	Less expensive, easier to install, and commonly used in everyday networking.	More expensive, requires specialized equipment and careful installation, but offers superior performance.

Real-world examples

- **Ethernet cable:** Connecting a **home Wi-Fi router** to a desktop computer or gaming console.
- **Fiber optic cable:** Connecting servers in a **data center** or serving as the **backbone network** across a college campus.